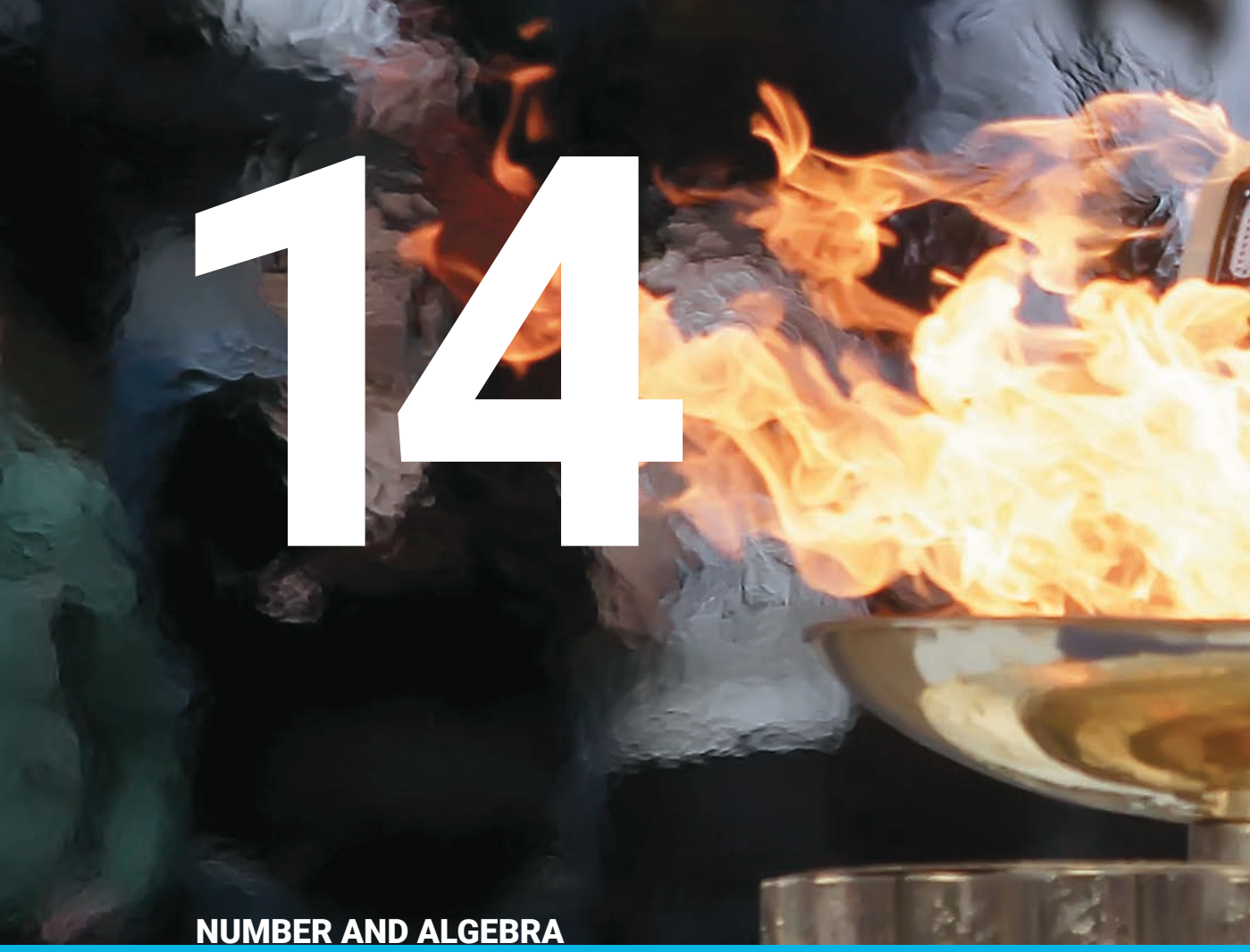


14

NUMBER AND ALGEBRA

QUADRATIC EQUATIONS AND THE PARABOLA

OPTIONAL PATH TOPIC
RECOMMENDED FOR STAGE 6 MATHEMATICS ADVANCED

Every 4 years, a few months before the Olympic Games Opening Ceremony, the Olympic flame is lit in a symbolic ceremony near the Temple of Hera, at the sacred site of the Ancient Olympics in Greece.

The flame is lit by traditional means, using the Sun's rays, to demonstrate its purity. To ensure it is lit by natural means, a parabolic mirror is used, which acts as a giant magnifying glass.

The Olympic flame has been used to light the Olympic cauldron for every Olympic Games Opening ceremony, since the Summer Olympic Games held in Berlin, Germany in 1936.


Ververidis Vasilis/Shutterstock.com

Chapter outline

		Working mathematically				
14.01	Quadratic equations $ax^2 + bx + c = 0$*	U	F	PS	R	C
14.02	Completing the square*	U	F		R	C
14.03	The quadratic formula*	U	F			C
14.04	Which method is best?*	U	F	PS	R	C
14.05	Graphing quadratic equations $y = ax^2 + bx + c$*	U	F		R	C
14.06	The axis of symmetry and vertex*	U	F		R	C
14.07	Non-linear simultaneous equations#	U	F		R	C

*PATH

Note: Chapter 12, *Products and factors*, must be completed before this chapter.

Wordbank

axis of symmetry The vertical line that divides a parabola in half and passes through the parabola's vertex

coefficient The number in front of a variable; for example, in $y = 3x^2 + 4x - 6$, the coefficient of x^2 is 3

quadratic equation An equation in which the highest power of the variable is 2; for example, $2x^2 - 12x + 10 = 0$

quadratic expression An algebraic expression in which the highest power of the variable is 2; for example, $3x^2 + 4x - 6$

quadratic formula A formula for solving quadratic equations of the form $ax^2 + bx + c = 0$

solution The answer to an equation, inequality or problem, the correct value(s) of the variable that makes an equation or inequality true

vertex A turning point of a curve; a parabola has one vertex

***x*-intercept** The *x*-value(s) at which a graph intersects the *x*-axis

Quiz
Wordbank 14

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communication

Videos (5):
14.01 Quadratic equations by factorising
14.02 Solving quadratic equations
• Simple quadratic equations
14.03 The quadratic formula
• Solving quadratic equations
14.07 Non-linear simultaneous equations

Quizzes (2):
- Wordbank 14
- SkillCheck 14

Worksheets (5):
14.03 Investigating quadratic equations
14.05 Graphing parabolas 2
14.06 Features of a parabola
• A page of parabolas • Parabolas

Puzzles (4):
14.01 Factorising quadratic equations
14.02 Complete the square order activity
14.03 Quadratic equations puzzle
• The quadratic formula

Technology (2):
14.03 The quadratic formula
14.05 Investigating parabolas 1

Spreadsheet (2):
14.03 The quadratic formula
14.05 Investigating parabolas 1

Nelson MindTap

To access resources above, visit
cengage.com.au/nelsonmindtap

In this chapter you will:

- (PATH) solve quadratic equations of the form $ax^2 + bx + c = 0$ by factorising, completing the square and using the quadratic formula
- (PATH) solve problems involving quadratic equations
- (PATH) graph parabolas of the form $y = ax^2 + bx + c$ and identify their features
- (PATH) find the axis of symmetry and the vertex of a parabola
- (PATH) find the point of intersection of a line with a parabola or hyperbola by solving linear and non-linear simultaneous equations

PATH

Quiz
SkillCheck 14

SkillCheck ANSWERS ON P. 638

1 Solve each quadratic equation.

a $4x^2 = 100$ **b** $3m^2 = 12$ **c** $x^2 - 81 = 0$
d $12k^2 = 3$ **e** $4u^2 - 49 = 0$ **f** $7w^2 - 63 = 0$

2 Factorise each expression.

a $16 - m^2$ **b** $d^2 - 121$ **c** $14y - 2y^2$
d $10p^2 + 25p$ **e** $5x^2 - 320$ **f** $18w^2 - 50$
g $k^2 + 5k + 4$ **h** $y^2 - 10y + 16$ **i** $m^2 - m - 56$
j $u^2 + 8u - 65$ **k** $w^2 - 10w + 21$ **l** $x^2 - 2x - 24$

3 If $y = x^2 + 4x - 7$, find the value of y if:

a $x = 1$ **b** $x = -1$ **c** $x = 2$ **d** $x = -\frac{1}{2}$

4 Factorise each expression.

a $3a^2 + 10a + 3$ **b** $5x^2 - 13x - 6$ **c** $6y^2 + y - 40$

d $15t^2 + 7t - 4$ **e** $5v^2 - 32v - 21$ **f** $8y^2 + 34y + 35$

g $15h^2 - 23h + 4$ **h** $12p^2 + 11p - 15$ **i** $16d^2 + 40d + 25$

PATH

14.01

Quadratic equations $ax^2 + bx + c = 0$ 14.01

In Chapter 6, *Equations and inequalities*, we learnt that in a **quadratic equation**, the highest power of the variable is 2; for example, $x^2 = 5$, $3m^2 + 7 = 10$, $d^2 - d - 6 = 0$ and $4y^2 - 3y = 8$.

The general form of a quadratic equation is $\boldsymbol{ax^2 + bx + c = 0}$, where a, b and c are constants (numbers) and $a \neq 0$.

Solving a quadratic equation means finding those values of the variable that satisfy the equation (or make the statement true). When checking solutions, substitute the possible solutions into the equation and show that the left-hand side (LHS) of the equation is equal to the right-hand side (RHS) of the equation.

PATH

Puzzle
Factorising quadratic equations

Solving $ax^2 + bx + c = 0$ by factorising

To solve quadratic equations of the form $ax^2 + bx + c = 0$, we need to factorise the quadratic expression on the LHS (see Chapter 12, *Products and factors*).

Example 1

Solve each quadratic equation.

a $4w(3w + 2) = 0$ **b** $(5m + 2)(2m - 7) = 0$

Solution

a $4w(3w + 2) = 0$

$\therefore 4w = 0$ or $3w + 2 = 0$

$w = 0$ or $3w = -2$

$w = 0$ or $w = -\frac{2}{3}$

b $(5m + 2)(2m - 7) = 0$

$5m + 2 = 0$ or $2m - 7 = 0$

$5m = -2$ or $2m = 7$

$m = -\frac{2}{5}$ or $m = \frac{7}{2}$

$m = -\frac{2}{5}$ or $m = 3\frac{1}{2}$

PATH

Video
Quadratic equations by factorising

Example 2

Solve each quadratic equation. Use substitution to check your solutions.

a $6p^2 - 9p = 0$ **b** $2x^2 - x - 15 = 0$ **c** $6y^2 = -7y + 5$

Solution

a $6p^2 - 9p = 0$ Factorising

$3p(2p - 3) = 0$

$3p = 0$ or $2p - 3 = 0$

$p = 0$ or $2p = 3$

$p = 0$ or $p = \frac{3}{2}$

$p = 0$ or $p = 1\frac{1}{2}$

Check: When $p = 0$, LHS $= 6 \times 0^2 - 9 \times 0 = 0 =$ RHS

When $p = 1\frac{1}{2}$, LHS $= 6 \times \left(1\frac{1}{2}\right)^2 - 9 \times 1\frac{1}{2} = 0 =$ RHS

b $2x^2 - x - 15 = 0$

$2x^2 - 6x + 5x - 15 = 0$ Product $= -30$, sum $= -1$

$2x(x - 3) + 5(x - 3) = 0$

$(x - 3)(2x + 5) = 0$

$x - 3 = 0$ or $2x + 5 = 0$

$x = 3$ or $2x = -5$

$x = 3$ or $x = -\frac{5}{2}$

$x = 3$ or $x = -2\frac{1}{2}$

Check: When $x = 3$, LHS $= 2 \times 3^2 - 3 - 15 = 0 =$ RHS

When $x = -2\frac{1}{2}$, LHS $= 2 \times \left(-2\frac{1}{2}\right)^2 - \left(-2\frac{1}{2}\right) - 15 = 0 =$ RHS

c $6y^2 = -7y + 5$

$6y^2 + 7y - 5 = 0$ Move RHS to LHS to make '$= 0$'

$6y^2 - 3y + 10y - 5 = 0$ Product $= -30$, sum $= 7$

$3y(2y - 1) + 5(2y - 1) = 0$

$(2y - 1)(3y + 5) = 0$

$2y - 1 = 0$ or $3y + 5 = 0$

$2y = 1$ or $3y = -5$

$y = \frac{1}{2}$ or $y = -\frac{5}{3}$

$y = \frac{1}{2}$ or $y = -1\frac{2}{3}$

Check: When $y=\frac{1}{2}$:

$\text{LHS}=6\times\left(\frac{1}{2}\right)^2=1\frac{1}{2}$

$\text{RHS}=-7\times\left(\frac{1}{2}\right)+5=1\frac{1}{2}$

LHS = RHS

When $y=-1\frac{2}{3}$:

$\text{LHS}=6\times\left(-1\frac{2}{3}\right)^2=16\frac{2}{3}$

$\text{RHS}=-7\times\left(-1\frac{2}{3}\right)+5=16\frac{2}{3}$

LHS = RHS

PATH

14.01

EXERCISE 14.01 ANSWERS ON P. 639

Quadratic equations $ax^2 + bx + c = 0$ U F PS R C

1 Solve each quadratic equation. EXAMPLE 1

a $(m+7)(m+3)=0$ **b** $(d-3)(d-7)=0$ **c** $(y-3)(y+5)=0$

d $k(k-3)=0$ **e** $t(t+7)=0$ **f** $2p(p-3)=0$

g $w(3w-2)=0$ **h** $(2n+1)(n-3)=0$ **i** $(5a-3)(2a-1)=0$

j $(3x+1)(2x+3)=0$ **k** $(2c-5)^2=0$ **l** $(1-2f)^2=0$

m $(3c+1)(4c+1)=0$ **n** $(1-2h)(h+1)=0$ **o** $(5-7e)(1-e)=0$

2 Solve each quadratic equation. EXAMPLE 2

a $2y^2+7y+6=0$ **b** $2g^2+5g+3=0$ **c** $3d^2+5d+2=0$

d $5t^2+16t+11=0$ **e** $2p^2-11p+12=0$ **f** $10x^2-19x+6=0$

g $8y^2-2y-3=0$ **h** $6a^2-5a-4=0$ **i** $4w^2-7w-15=0$

j $5c^2+2c-7=0$ **k** $8e^2+10e-3=0$ **l** $3q^2+4q-15=0$

m $4g^2-20g+25=0$ **n** $18m^2-3m-10=0$ **o** $16-8w-3w^2=0$

p $36+3y-3y^2=0$ **q** $2f^2-24f+72=0$ **r** $12h^2+3h-9=0$

3 Express each quadratic equation in the form $ax^2+bx+c=0$ and solve.

a $2x^2=x+15$ **b** $4t(t+2)=5$ **c** $41u=-8u^2-5$

d $7m^2=8m-1$ **e** $p(p-3)=28$ **f** $(e-2)^2=9$

g $t(2t-13)=-15$ **h** $7=6d^2+11d$ **i** $5h^2=125$

j $8f^2=4f$ **k** $6w^2+3=19w$ **l** $4a(3a+5)-8=0$

4 A certain positive number, plus its square, minus 72, equals 0.
Find the number. PS R C

Foundation Standard Complex

14.02 Completing the square

PATH

Video
Solving quadratic equations

Puzzle
Complete the square order activity

Quadratic equations can be solved by using a method called **completing the square**. We try to make the LHS of the equation a perfect square (see Chapter 12, *Products and factors*).

The method of **completing the square** is based on the following results for perfect squares.

$(x + a)^2 = x^2 + 2ax + a^2$

$(x - a)^2 = x^2 - 2ax + a^2$

We note that the last term, a^2, is the square of 'half the coefficient of x'.

Example 3

Find the numbers that complete the square in each equation.

a $x^2 + 10x + ... = (x + ...)^2$

b $x^2 - 14x + ... = (x - ...)^2$

Solution

a The coefficient of x is 10.
Half of 10 is 5, and $5^2 = 25$.
The perfect square is
$x^2 + 10x + 25 = (x + 5)^2$

b The coefficient of x is -14.
Half of -14 is -7, and $(-7)^2 = 49$.
The perfect square is
$x^2 - 14x + 49 = (x - 7)^2$

Video
Simple quadratic equations

Example 4

Solve $(k + 3)^2 = 7$.

Solution

$(k + 3)^2 = 7$

$k + 3 = \pm\sqrt{7}$ Taking the square root of both sides.

$k + 3 - 3 = \sqrt{7} - 3$ **or** $k + 3 - 3 = -\sqrt{7} - 3$

This solution is usually written as:

$k = -3 + \sqrt{7}$ **or** $k = -3 - \sqrt{7}$

Notice that the 2 solutions are surds (not rational), of the form $a + \sqrt{b}$ and $a - \sqrt{b}$.

Example 5

Solve $x^2 + 6x - 2 = 0$ by completing the square.

Solution

Step 1

Move the constant term to the RHS.

$x^2 + 6x = 2$

Step 2

Halve the coefficient of x, square it and then add the square to both sides.

$x^2 + 6x + 3^2 = 2 + 3^2$

$x^2 + 6x + 9 = 11$

Step 3

Express the LHS as a perfect square.

$(x + 3)^2 = 11$

Step 4

Solve the resulting equation.

$x + 3 = \pm\sqrt{11}$ — Taking the square root of both sides.

$x = -3 \pm \sqrt{11}$ — This answer in surd form is called the 'exact answer'.

$x = -3 + \sqrt{11}$ or $x = -3 - \sqrt{11}$

Example 6

Solve $2x^2 - 3x - 4 = 0$, writing the solution correct to 2 decimal places.

Solution

$2x^2 - 3x - 4 = 0$

$2x^2 - 3x = 4$ — Moving the constant term to the RHS.

$x^2 - \frac{3}{2}x = 2$ — Dividing both sides by 2, the coefficient of x^2.

$x^2 - \frac{3}{2}x + \left(-\frac{3}{4}\right)^2 = 2 + \left(-\frac{3}{4}\right)^2$ — Completing the square.

$\left(x - \frac{3}{4}\right)^2 = \frac{41}{16}$ — Expressing the LHS as a perfect square and leaving the RHS as an improper fraction.

$x - \frac{3}{4} = \pm\sqrt{\frac{41}{16}}$ — Taking the square root of both sides.

$x - \frac{3}{4} = \pm\frac{\sqrt{41}}{4}$ — Simplifying the RHS.

$x = \frac{3}{4} \pm \frac{\sqrt{41}}{4}$ — This is the exact answer in surd form.

$x = \frac{3 \pm \sqrt{41}}{4}$

$x = \frac{3+\sqrt{41}}{4}$ or $x = \frac{3-\sqrt{41}}{4}$ — This is an approximate answer in decimal form.

$x = 2.35078...$ or $x = -0.85078...$

$x \approx 2.35$ or $x \approx -0.85$

PATH

EXERCISE 14.02 ANSWERS ON P. 639

Completing the square U F R C

EXAMPLE 3

1 Find the numbers that 'complete the square' in each equation. R C

a $x^2 + 2x + ___ = (x + ___)^2$

b $p^2 - 6p + ___ = (p - ___)^2$

c $m^2 - 8m + ___ = (m - ___)^2$

d $k^2 + 4k + ___ = (k + ___)^2$

e $y^2 - 7y + ___ = (y - ___)^2$

f $w^2 - 3w + ___ = (w - ___)$

g $x^2 + x + ___ = (x + ___)^2$

h $h^2 - 5h + ___ = (h - ___)^2$

i $a^2 + \frac{7}{2}a + ___ = (a + ___)^2$

j $v^2 + \frac{5}{3}v + ___ = (v + ___)^2$

EXAMPLE 4

2 Solve each equation, writing the solution in surd form. R C

a $(d + 3)^2 = 7$

b $(x - 5)^2 = 5$

c $(p + 1)^2 = 10$

d $(y - 1)^2 = 2$

e $\left(m - \frac{1}{2}\right)^2 = 5$

f $\left(t + \frac{2}{3}\right)^2 = 3$

g $(c + 1)^2 = \frac{21}{2}$

h $(w - 3)^2 = \frac{41}{2}$

i $\left(n + \frac{2}{3}\right)^2 = \frac{7}{9}$

j $\left(e - \frac{3}{2}\right)^2 = \frac{71}{4}$

k $(d - 2)^2 = 5$

l $\left(x - \frac{3}{4}\right)^2 = 2$

EXAMPLE 5

3 Solve each equation by completing the square. Leave your answers in exact form. R C

a $h^2 + 2h - 5 = 0$

b $r^2 - 2r - 1 = 0$

c $m^2 + 6m + 2 = 0$

d $w^2 - 4w - 60 = 0$

e $a^2 - 10a - 5 = 0$

f $y^2 + 8y - 3 = 0$

g $p^2 + 12p - 5 = 0$

h $x^2 - 4x + 2 = 0$

i $u^2 + 9u + 14 = 0$

j $d^2 + d - 7 = 0$

k $c^2 - 9c + 2 = 0$

l $e^2 + 5e + 2 = 0$

m $y^2 - 3y - 8 = 0$

n $b^2 - b - 5 = 0$

o $q^2 - 3q + 1 = 0$

p $2g^2 + 7g = 3$

q $2x^2 = 7 - 5x$

r $3f(f + 4) = 6$

EXAMPLE 6

4 Solve each of these by completing the square. Give your answers correct to 2 decimal places if irrational. R C

a $x^2 + 12x + 9 = 0$

b $m^2 - 16m - 7 = 0$

c $g^2 + 4g - 3 = 0$

d $2h^2 + 3h - 7 = 0$

e $5w^2 - 4w - 3 = 0$

f $3y^2 + y - 5 = 0$

g $p(3p + 2) = 8$

h $4e^2 - 4 = e$

i $2n^2 = 5 - 3n$

The quadratic formula

14.03

There is a formula for solving a quadratic equation of the form $ax^2 + bx + c = 0$ that involves the coefficients a, b and c.

The quadratic formula

The solutions to the quadratic equation $ax^2 + bx + c = 0$ are:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Example 7

Solve each quadratic equation using the quadratic formula.

a $6x^2 + x - 2 = 0$ **b** $3x^2 + 11x + 2 = 0$

Solution

a For $6x^2 + x - 2 = 0$:

$a = 6$, $b = 1$, and $c = -2$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
$$= \frac{-1 \pm \sqrt{1^2 - 4 \times 6 \times (-2)}}{2 \times 6}$$
$$= \frac{-1 \pm \sqrt{49}}{12}$$
$$= \frac{-1 \pm 7}{12}$$

$x = \frac{-1+7}{12}$ or $x = \frac{-1-7}{12}$

$x = \frac{6}{12}$ or $x = \frac{-8}{12}$

$x = \frac{1}{2}$ or $x = -\frac{2}{3}$

b For $3x^2 + 11x + 2 = 0$:

$a = 3$, $b = 11$ and $c = 2$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
$$= \frac{-11 \pm \sqrt{11^2 - 4 \times 3 \times 2}}{2 \times 3}$$
$$= \frac{-11 \pm \sqrt{97}}{6}$$

$x = \frac{-11 \pm \sqrt{97}}{6}$ or $x = \frac{-11 - \sqrt{97}}{6}$ In exact surd form

PATH

14.03

Worksheet
Investigating quadratic equations

Puzzles
Quadratic equations puzzle

The quadratic formula

Technology
The quadratic formula

Spreadsheet
The quadratic formula

Videos
The quadratic formula

Solving quadratic equations

PATH

Example 8

Solve $2x^2 - 6 = 9x$ using the quadratic formula, expressing the answer correct to 2 decimal places.

Solution

$2x^2 - 6 = 9x$

$2x^2 - 9x - 6 = 0$

Writing the quadratic equation in the form $ax^2 + bx + c = 0$

Substitute $a = 2, b = -9, c = -6$

$$x = \frac{-(-9) \pm \sqrt{(-9)^2 - 4 \times 2 \times (-6)}}{2 \times 2}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{9 \pm \sqrt{129}}{4}$$

$x = \frac{9 + \sqrt{129}}{4}$ or $x = \frac{9 - \sqrt{129}}{4}$

$x = 5.0894...$ or $x = -0.5894...$

$x \approx 5.09$ or $x \approx -0.59$

EXERCISE 14.03 ANSWERS ON P. 640

The quadratic formula U F C

EXAMPLE 7

1 Solve each quadratic equation using the quadratic formula. Write each solution in exact form. C

a	$x^2 + 6x + 2 = 0$	**b**	$m^2 - 5m - 3 = 0$	**c**	$w^2 - 8w + 3 = 0$
d	$k^2 + 3k - 5 = 0$	**e**	$y^2 - 4y - 1 = 0$	**f**	$p^2 + p - 5 = 0$
g	$u^2 - 7u - 3 = 0$	**h**	$2a^2 + 3a - 7 = 0$	**i**	$5q^2 - 6q + 1 = 0$
j	$3c^2 + 2c - 2 = 0$	**k**	$4e^2 - 5e - 2 = 0$	**l**	$3x^2 + 8x + 2 = 0$
m	$2d^2 - 4d - 5 = 0$	**n**	$3a^2 - 10a - 2 = 0$	**o**	$2t^2 + 3t - 5 = 0$
p	$3y^2 + 8y + 4 = 0$	**q**	$6k^2 - 11k + 5 = 0$	**r**	$2n^2 - 5n - 11 = 0$

2 Solve each quadratic equation using the quadratic formula, expressing the solution as a surd. C

a	$5y^2 - 9y = 3$	**b**	$3m^2 = 7 - 2m$	**c**	$4x^2 = 3x + 2$
d	$1 - 4k - k^2 = 0$	**e**	$3m^2 - 1 - 3m = 0$	**f**	$1 - 2g - 5g^2 = 0$
g	$8 = 9h - 2h^2$	**h**	$2w + 2 = 3w^2$	**i**	$4p - 3p^2 = -1$
j	$2 - 4u - 5u^2 = 0$	**k**	$6a^2 = 9 - 4a$	**l**	$10 = 3y + 2y^2$

3 Solve each quadratic equation, writing the solutions correct to 2 decimal places. C

a	$k^2 - 9k + 1 = 0$	**b**	$c^2 - 2 = 0$	**c**	$m^2 - 5 = 2$
d	$2n^2 + 2 = 7n$	**e**	$2p^2 + 3p - 4 = 0$	**f**	$6w^2 + 5w - 2 = 0$
g	$3x^2 + 2 - 8x = 0$	**h**	$h^2 = 7 + 2h$	**i**	$1 + x - x^2 = 0$
j	$36 = 17a - a^2$	**k**	$5v^2 - 11 = 0$	**l**	$5c^2 + 8 = 15c$
m	$t^2 = 5(t + 5)$	**n**	$(x - 6)^2 = 3$	**o**	$12 = 2d^2 - 3d$

PATH

14.04

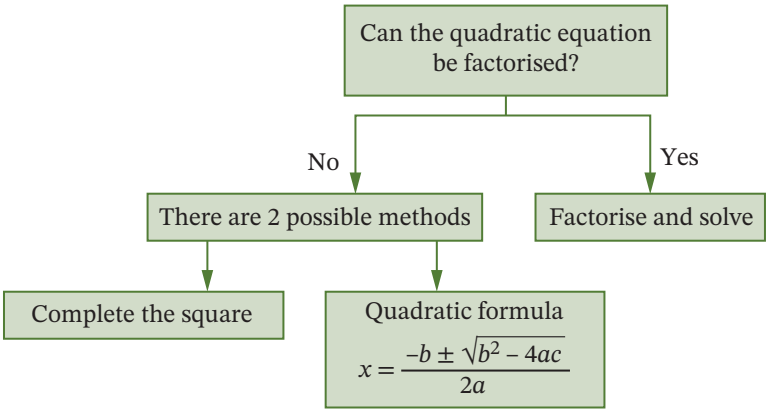
EXAMPLE 8

Which method is best?

14.04

We have learned 3 methods of solving quadratic equations.

Factorising is the simplest, but only some quadratic equations can be factorised, while the **quadratic formula** works for all quadratic equations. **Completing the square** also works for all equations, but the algebra can be messy.



PATH

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PATH

EXERCISE 14.04 ANSWERS ON P. 640

Which method is best? U F R

1 Solve each quadratic equation using the most suitable method. You may want to try 2 methods to compare your answers and determine how effective the methods were. R

a $x(2x - 3) = 0$ **b** $m^2 = 36$ **c** $3w^2 - 5w - 1 = 0$

d $u^2 - u - 12 = 0$ **e** $(5k - 3)(2k + 7) = 0$ **f** $16 = 25a^2$

g $x(x - 2) = 63$ **h** $n^2 - 6n = 10$ **i** $(a + 3)^2 = 5$

j $(d + 3)(d - 1) = 0$ **k** $t^2 = 4t$ **l** $(2y - 3)^2 = 11$

m $30 - x^2 = 7x$ **n** $6a^2 = 6 - 5a$ **o** $4x = 12x^2 - 3$

2 Solve each quadratic equation using an appropriate method, giving your answers correct to 2 decimal places where necessary. R

a $y^2 = 5y$ **b** $144 = a^2$ **c** $(u - 2)^2 = 11$

d $x^2 - 6x = -9$ **e** $18 + 7x - x^2 = 0$ **f** $x^2 + 10x + 5 = 0$

g $a(2a - 1) = 7$ **h** $2w^2 = 3w + 7$ **i** $(5m - 1)^2 = 18$

j $(3 - 2k)^2 = 27$ **k** $\frac{36}{x} + 1 = 2x$ **l** $\frac{h}{3h + 2} = \frac{h - 3}{4h}$

14.05 Graphing quadratic equations $y = ax^2 + bx + c$

PATH

Worksheet Graphing parabolas 2

Technology Investigating parabolas 1

Spreadsheet Investigating parabolas 1

The graph of a quadratic equation is a smooth U-shaped curve called a **parabola**. We have already graphed simple parabolas of the form $y = ax^2 + c$ in Chapter 7, *Graphing curves*.

Now we will graph parabolas of the form $y = ax^2 + bx + c$.

Example 9

For each quadratic equation:

i complete a table of values and draw the graph

ii find the x-intercepts

iii find the y-intercept

iv solve $y = 0$ and compare the solutions to the x-intercepts.

a $y = x^2 + 6x + 5$ **b** $y = 2x^2 - x - 10$

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Solution

a **i** $y = x^2 + 6x + 5$

ii The x-intercepts are -5 and -1.

iii The y-intercept is 5.

Note that this is the constant term, $c = 5$, in $y = x^2 + 6x + 5$.

x	−6	−5	−4	−3	−2	−1	0	1
y	5	0	−3	−4	−3	0	5	12

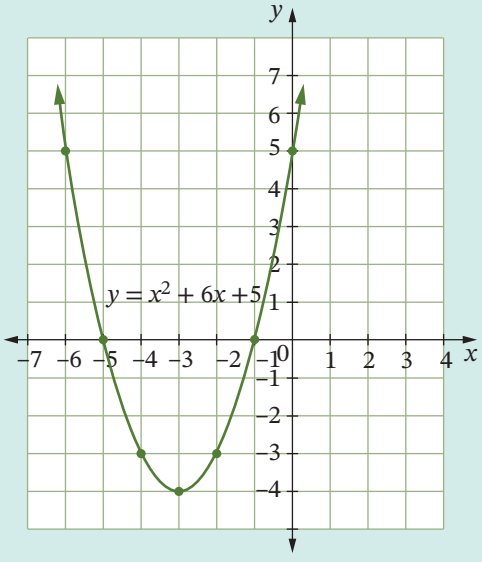
iv $x^2 + 6x + 5 = 0$

$(x + 5)(x + 1) = 0$

$x + 5 = 0$ or $x + 1 = 0$

$x = -5$ or $x = -1$

The solutions to the quadratic equation are the same as the x-intercepts of the graph of $y = x^2 + 6x + 5$.



b **i** $y = 2x^2 - x - 10$

ii The x-intercepts are -2 and $2\frac{1}{2}$.

iii The y-intercept is -10.

Note that this is the constant term, $c = -10$, in $y = 2x^2 - x - 10$.

x	−3	−2	−1	0	1	2	3
y	11	0	−7	−10	−9	−4	5

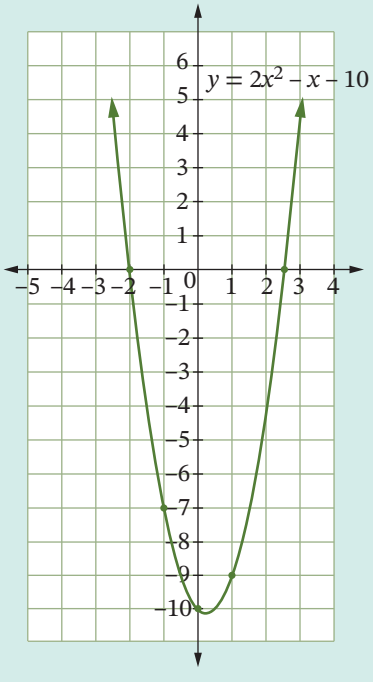
iv $2x^2 - x - 10 = 0$

$(2x - 5)(x + 2) = 0$

$2x - 5 = 0$ or $x + 2 = 0$

$x = 2\frac{1}{2}$ or $x = -2$

The solutions to the quadratic equation are the same as the x-intercepts of the graph of $y = 2x^2 - x - 10$.



PATH

The graph of the parabola $y = ax^2 + bx + c$

- If $a > 0$ (positive), the parabola is **concave up**
- If $a < 0$ (negative), the parabola is **concave down**
- The ***y*-intercept** is where $x = 0$, so it is the value of c
- The ***x*-intercepts** are where $y = 0$, so are the solutions to the quadratic equation $ax^2 + bx + c = 0$
- Because the parabola is symmetrical, the x-coordinate of its vertex is halfway between the x-intercepts.

Example 10

Graph each quadratic equation, showing its x- and y-intercepts.

a $y = x(2x - 5)$ **b** $y = 2x^2 + x - 6$ **c** $y = -2x^2 - 3x + 9$

Solution

a $y = x(2x - 5)$

$= 2x^2 - 5x$

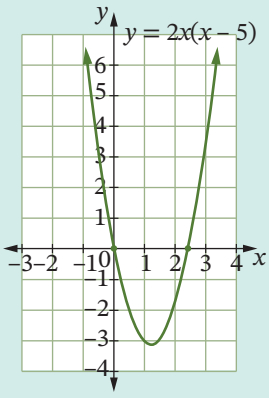
$a = 2 > 0$, so the parabola is concave up

x-intercepts: $x(2x - 5) = 0$

$x = 0$ and $2x - 5 = 0$

$x = 0$ and $x = 2\frac{1}{2}$

y-intercept: $c = 0$



b $y = 2x^2 + x - 6$

$a = 2 > 0$, so the parabola is concave up

x-intercepts:

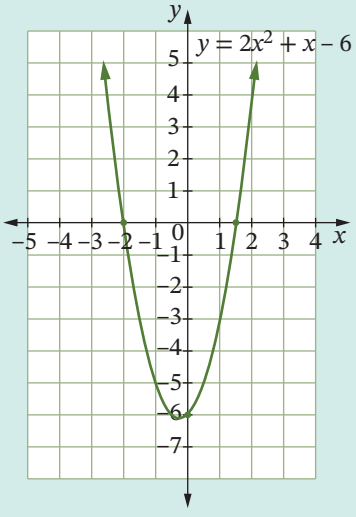
$2x^2 + x - 6 = 0$

$(2x - 3)(x + 2) = 0$

$2x - 3 = 0$ and $x + 2 = 0$

$x = 1\frac{1}{2}$ and $x = -2$

y-intercept: $c = -6$



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c $y = -2x^2 - 3x + 9$

$a = -2 < 0$, so the parabola is concave down

x-intercepts:

$-2x^2 - 3x + 9 = 0$ Dividing by -1

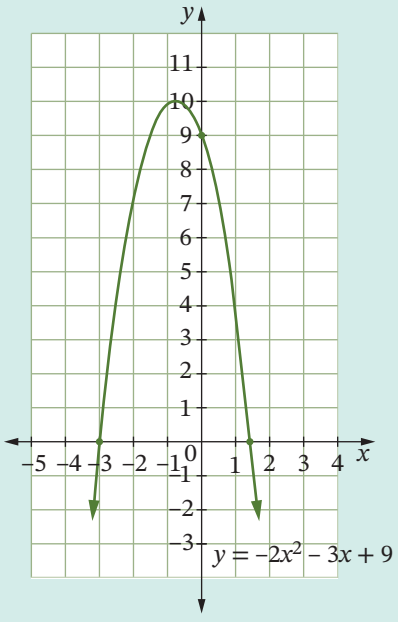
$2x^2 + 3x - 9 = 0$

$(2x - 3)(x + 3) = 0$

$2x - 3 = 0$ and $x + 3 = 0$

$x = 1\frac{1}{2}$ and $x = -3$

y-intercept: $c = 9$



EXERCISE 14.05 ANSWERS ON P. 640

Graphing quadratic equations $y = ax^2 + bx + c$ U F R C

1 For each quadratic equation: R C

- **i** complete a table of values and draw the graph
- **ii** find the x-intercepts
- **iii** find the y-intercept
- **iv** solve $y = 0$ and compare the solutions to the x-intercepts.

a $y = x^2 + 4x + 3$ **b** $y = 2x^2 + 7x + 3$

c $y = 2x^2 - 3x - 9$ **d** $y = x^2 - 2x$

EXAMPLE 9

2 Find the y-intercept of the parabolas with equation:

a $y = 3x^2 - 2x - 5$ **b** $y = 2x^2 + 6x + 3$ **c** $y = 5x^2 - 10x$.

3 Graph each quadratic equation, showing its x- and y-intercepts.

EXAMPLE 10

a $y = x(x - 4)$ **b** $y = 3x(x + 2)$ **c** $y = (x + 3)(x - 5)$

d $y = (x - 2)(2x - 5)$ **e** $y = -(x + 1)(3x + 2)$ **f** $y = x^2 + 6x + 8$

g $y = -x^2 + 4x + 5$ **h** $y = 3x^2 + 11x - 20$ **i** $y = -2x^2 + 5x - 2$

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14.06 The axis of symmetry and vertex

PATH

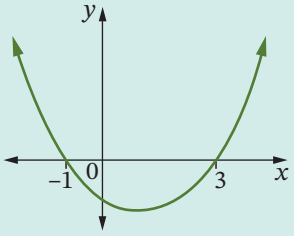
Worksheets
Features of a parabola
A page of parabolas
Parabolas

The axis of symmetry and vertex of $y = ax^2 + bx + c$

- The **axis of symmetry** of the parabola is the vertical line: $x = -\frac{b}{2a}$
- The **axis of symmetry** passes through the point halfway between the 2 x-intercepts of the parabola
- The **vertex** of the parabola lies on the axis of symmetry, so its ***x*-coordinate** is $x = -\frac{b}{2a}$ and its ***y*-coordinate** can be found by substituting the x-coordinate into the equation of the parabola

Example 11

Find the equation of the axis of symmetry of this parabola.



Solution

The x-intercepts are -1 and 3.

The equation of the axis of symmetry is:

$$x = \frac{-1+3}{2}$$
$$= 1$$

Finding the average of the x-intercepts.

Example 12

A parabola has the equation $y = 2x^2 - 4x + 3$.

a Find the equation of its axis of symmetry

b Find the coordinates of the vertex of the parabola.

Solution

a For $y = 2x^2 - 4x + 3$, $a = 2$, $b = -4$ and $c = 3$.

The axis of symmetry is:

$$x = -\frac{b}{2a}$$
$$= -\frac{-4}{2 \times 2}$$
$$= 1$$

$x = 1$ is the equation of the axis of symmetry.

b The vertex lies on the axis of symmetry.

Substitute $x = 1$ in $y = 2x^2 - 4x + 3$

$y = 2 \times 1^2 - 4 \times 1 + 3$

$= 1$

The vertex is (1, 1).

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Example 13

Sketch the parabola $y = 2x^2 + 5x + 1$.

Solution

$a = 2 > 0$, so the parabola is concave up.

y-intercept $= 1$.

$y = 2x^2 + 5x + 1$ cannot be factorised, so we cannot find the x-intercepts precisely.

For the vertex:

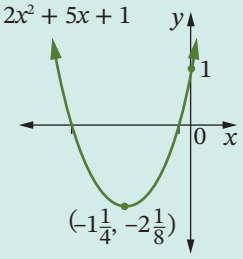
$$x = -\frac{b}{2a} \quad \text{where } a = 2, \text{ and } b = 5$$
$$= -\frac{5}{2 \times 2}$$
$$= -1\frac{1}{4}$$

Substitute $x = -1\frac{1}{4}$ into $y = 2x^2 + 5x + 1$

$$y = 2 \times \left(-1\frac{1}{4}\right)^2 + 5 \times \left(-1\frac{1}{4}\right) + 1$$
$$= -2\frac{1}{8}$$

$\therefore$ The vertex is $\left(-1\frac{1}{4}, -2\frac{1}{8}\right)$.

$y = 2x^2 + 5x + 1$





JossK/Shutterstock.com

PATH

Graphing the parabola $y = ax^2 + bx + c$

- Use the sign of a (the coefficient of x^2) to determine whether the parabola is **concave up** or **down**
- Find the y-**intercept**, c
- Find the x-**intercepts** where possible by solving a quadratic equation
- Use $x = -\frac{b}{2a}$ to find the **axis of symmetry** and the **vertex**

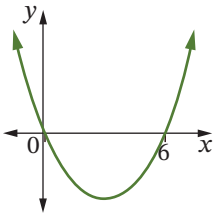
EXERCISE 14.06 ANSWERS ON P. 642

The axis of symmetry and vertex UFRC

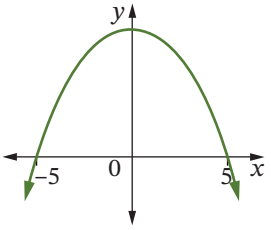
EXAMPLE 11

1 Write the equation for the axis of symmetry in each parabola. R C

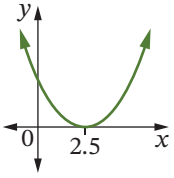
a



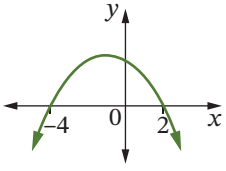
b



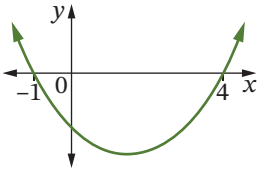
c



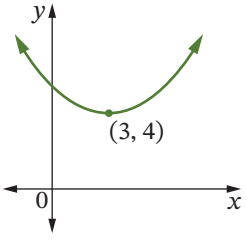
d



e



f



EXAMPLE 12

2 For each parabola with the equation given below:

i find the equation of its axis of symmetry

ii find the coordinates of its vertex.

a	$y = x^2 - 6x + 8$	**b**	$y = -x^2 + 10x - 9$	**c**	$y = x^2 - 2x + 10$
d	$y = -x^2 + 8x + 9$	**e**	$y = -x^2 + x - 25$	**f**	$y = 5x^2 + 40x$
g	$y = 24x - 3x^2$	**h**	$y = 4x^2 + 2x - 1$	**i**	$y = 1 - 3x - 9x^2$

EXAMPLE 13

3 Graph each quadratic equation, showing:

i the x-intercepts (correct to one decimal place where necessary)

ii the y-intercept

iii the equation of the axis of symmetry

iv the coordinates of the vertex

v the concavity.

a	$y = x^2 - 6x - 40$	**b**	$y = x^2 - 3x$	**c**	$y = 2x^2 + 3x + 4$
d	$y = -x^2 + 6x + 5$	**e**	$y = -4x^2 - 12x + 21$	**f**	$y = x^2 - 8x + 3$
g	$y = 5x^2 + 7x + 4$	**h**	$y = 8x - 2x^2$	**i**	$y = -2x^2 + 7x - 6$

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Non-linear simultaneous equations

14.07

The points of intersection of a line with a parabola, circle or hyperbola may be found either graphically or algebraically. The algebraic method involves solving simultaneous equations using the substitution method, which is covered in Chapter 11, *Simultaneous equations.*

PATH

14.07

Video
Non-linear simultaneous equations

Example 14

Find the points of intersection of:

a the line $y = 2x - 3$ and the parabola $y = x^2 - 3x + 1$

b the line $y = x + 2$ and the circle $x^2 + y^2 = 4$

c the line $y = x + 5$ and the hyperbola $y = \frac{6}{x}$.

Solution

a $y = 2x - 3$ [1]

$y = x^2 - 3x + 1$ [2]

Use [2] to substitute for y in [1].

$x^2 - 3x + 1 = 2x - 3$

$x^2 - 5x + 4 = 0$

$(x - 4)(x - 1) = 0$

$x - 4 = 0$ or $x - 1 = 0$

$x = 4$ or $x = 1$

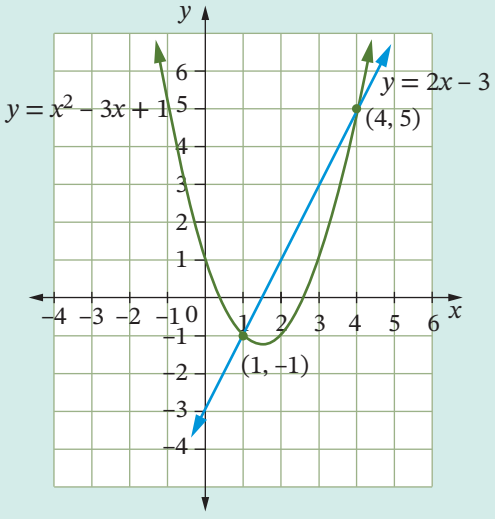
Substitute $x = 4$ and $x = 1$ into [1] to find y.

$y = 2 \times 4 - 3$ and $y = 2 \times 1 - 3$

$= 5$ $= -1$

The solutions are $x = 4, y = 5$ and $x = 1, y = -1$

The points of intersection are (4, 5) and (1, −1).



This solution can also be found by graphing both equations and finding their points of intersection, as shown in the graph.

PATH

b $y = x + 2$ [1]

$x^2 + y^2 = 4$ [2]

Use [1] to substitute for y in [2].

$$x^2 + (x+2)^2 = 4$$
$$x^2 + x^2 + 4x + 4 = 4$$
$$2x^2 + 4x = 0$$
$$2x(x+2) = 0$$

$2x = 0$ or $x + 2 = 0$

$x = 0$ or $x = -2$

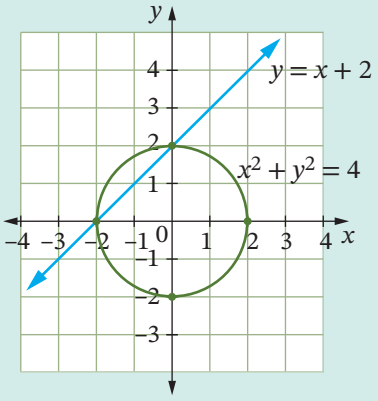
Substitute $x = 0$ and $x = -2$ into [1] to find y.

$y = 0 + 2$ and $y = -2 + 2$

$y = 2$ and $y = 0$

The points of intersection are $(0, 2)$ and $(-2, 0)$.

This solution can also be found by graphing both equations and finding their points of intersection, as shown in the graph.



c $y = x + 5$ [1]

$y = \frac{6}{x}$ [2]

Use [1] to substitute for y in [2].

$$x + 5 = \frac{6}{x}$$
$$x(x+5) = x \times \frac{6}{x}$$
$$x^2 + 5x = 6$$
$$x^2 + 5x - 6 = 0$$
$$(x+6)(x-1) = 0$$

$x + 6 = 0$ or $x - 1 = 0$

$x = -6$ or $x = 1$

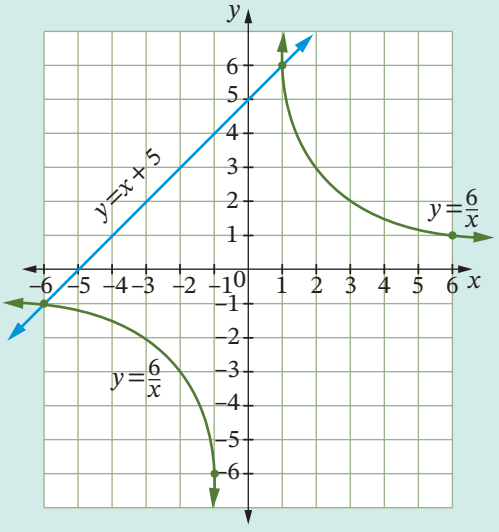
Substitute $x = -6$ and $x = 1$ into [1] to find y.

$y = -6 + 5$ and $y = 1 + 5$

$y = -1$ and $y = 6$

The points of intersection are $(-6, -1)$ and $(1, 6)$.

(Also, see their graphs).



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EXERCISE 14.07 ANSWERS ON P. 643

PATH

Non-linear simultaneous equations UFRC

EXAMPLE 14

14.07

1 Find the points of intersection of the line and the parabola with equations: R

a $y = x^2$ and $y = 5x - 6$

b $y = 2x^2$ and $y = 8x$

c $y = x^2 + 10$ and $y = 2x + 18$

d $y = 5x^2$ and $y = x + 6$

e $y = 3x^2 + 2x + 10$ and $y = 12 - 3x$

f $y = 9x - x^2$ and $y = 6x - 10$

g $y = 6x - 3$ and $y = x^2 - 8x + 46$

h $x + y = 2$ and $y = x^2 - 2x$

2 Find the points of intersection of the line: R

a $y = x - 3$ and the hyperbola $y = -\frac{2}{x}$

b $y = 1 - x$ and the circle $x^2 + y^2 = 1$

c $x + y = 3$ and the circle $x^2 + y^2 = 9$

d $y = x + 5$ and the circle $x^2 + y^2 = 25$

e $y = 5x + 2$ and the hyperbola $y = \frac{7}{x}$

f $2x - y = 1$ and the hyperbola $y = \frac{15}{x}$

g $y = 2x + 5$ and the hyperbola $y = -\frac{3}{x}$

3 Explain why there are no points of intersection between the equations $y = -3x - 1$ and $y = \frac{2}{x}$. Show working to help justify your answer. R C

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